Name: Naman Sahu

Registration Number: 25BMR10017

Course Code: CSA2001

Course: Fundamentals of AI and ML

Submission Date: 31 March 2026

Student Performance Predictor

1. Introduction

This project focuses on predicting student marks based on daily habits such as study hours, attendance, and sleep hours. The aim is to understand how these factors affect academic performance using basic machine learning techniques.

---

2. Problem Statement

Many students are unaware of how their daily routine impacts their academic results. This project aims to solve this problem by building a system that predicts marks based on key factors like study time, attendance, and sleep.

---

3. Objective

The main objectives of this project are:

- To build a simple machine learning model
- To analyze the relationship between daily habits and marks
- To provide predicted marks based on user input

---

4. Methodology

The project uses a supervised machine learning approach. A dataset containing student details is used to train a Linear Regression model.

Steps followed:

- Collected sample student data
- Loaded data using pandas
- Selected input features (study hours, attendance, sleep hours)
- Trained the model using Linear Regression
- Took user input from the terminal
- Predicted marks based on the input

---

5. Tools and Technologies Used

- Python
- Pandas
- Scikit-learn

---

6. Implementation

The project is implemented using Python. The dataset is stored in a CSV file. The model is trained using Linear Regression. After training, the program takes input from the user and predicts marks.

The entire project is structured into folders for better organization, including data, source code, and dependencies.

7. Results

The model successfully predicts student marks based on the given inputs. For example:

- Study hours: 6
- Attendance: 80
- Sleep hours: 7

Predicted Marks: around 70

8. Challenges Faced

- Understanding how to structure the project properly
- Handling file paths correctly
- Removing warnings during prediction
- Making the program interactive with user input

9. Learning Outcomes

- Learned basics of machine learning
- Understood Linear Regression
- Learned how to handle datasets using pandas
- Gained experience in building a complete project
- Learned how to use GitHub for project submission

10. Conclusion

This project demonstrates how machine learning can be used to solve real-world problems in a simple way. It shows that daily habits have a direct impact on student performance, and this can be predicted using basic models.

***